

Regulation functions by counting states

Four promoters of increasing architecture, and **one procedure applied to all four**, with less of my working shown each time. **We do items 1 and 2 in the room** — ten minutes, and item 1 is already done. Items 3 and 4 are on **PS3**.

The procedure, every time:

1	List every state the promoter can be in.	3	Say which states transcribe, and how fast.
2	Give each state a weight.	4	Divide the transcribing weight by the total.

Throughout: $a = [A]/K_A$ is an occupancy variable, p is the polymerase weight, and $f = e^{-\varepsilon_{ap}/k_B T}$ is charged **only** to a state in which two proteins are actually touching. Every answer below is taken in the **weak-promoter limit** ($p \ll 1$), where the fold-change is the regulation factor and p cancels.

1 • Fully worked — simple repression at a truncated lac promoter

lacUV5 cut down to a single operator O_m , with LacI₄ binding it. The operator **overlaps the polymerase site**: the repressor and RNAP cannot both be bound.

List the states. Three, not four. The missing one is what makes this repression.

STATE	WEIGHT	TRANSCRIBES?
empty	1	no
LacI ₄ on O_m	$r = [R]/K_m$	no
RNAP on the promoter	p	yes, at rate 1
<i>both</i>	—	does not exist

Divide the transcribing weight by the total.

$$p_{\text{bound}} = \frac{p}{1 + r + p}$$

Take the ratio to the unregulated promoter ($r = 0$), which is what an experiment reports:

$$\text{fold-change} = \frac{p/(1 + r + p)}{p/(1 + p)} = \frac{1 + p}{1 + r + p}$$

Take the weak-promoter limit, $p \ll 1$:

$$F_{\text{reg}} = \frac{1}{1 + [R]/K_m}$$

which is exactly the form you derived in session 4 from a binding equilibrium — reached here by counting states, with no $[P]$ to cancel.

► **Why does that step follow?** There is no f anywhere in this answer. Answer from the *mechanism*, not the algebra: what would f have been charged for, and what about this architecture means there is nothing to charge?

2 · The last step is yours — cooperative activation at λP_{RM}

Two operators. cl_2 at O_{R2} **activates** — it contacts polymerase, with enhancement factor f . cl_2 at O_{R1} does **not** contact polymerase at all; it helps recruit the other dimer, through a cooperative interaction ω between the two bound dimers. One protein species, so both occupancies move together: write $a = [cI_2]/K_{R2}$ and $h = [cI_2]/K_{R1}$.

List the states, and give each a weight. Four of them. Two are filled in.

STATE	WEIGHT	TRANSCRIBES AT
empty	1	1
cl_2 on O_{R2} only	a	f
cl_2 on O_{R1} only		
both		

► Where does ω go, and where does it not? One line, before you write the answer down.

Now divide, and take the weak-promoter limit.

$$F_{\text{reg}} =$$

Three checks, and do all three — they are worth more than the derivation.

► Set $\omega = 1$. What should the expression become, physically, and does it?

► Set $h = 0$ (delete O_{R1}). You should recover item 2 of the lecture.

► Let $[cI_2] \rightarrow \infty$. Show that the answer is f , whatever ω and K_{R1} are. This is one line, and it is the whole design result: the helper site cannot raise the ceiling.

3 · On PS3 — two different proteins at *melAB*

MeIR*₂ binds the weak proximal operator *O2* and activates. CRP*₂ binds the upstream operator *O1*, helps recruit MeIR, and **does not itself activate**.

Two species now, so the two occupancies move independently. Write the state list, the regulation factor, and then answer the question that matters: **whose *f* appears in the saturating fold-change, and why is there only one of them?**

4 · On PS3 — the design item

Design a promoter whose fold-change is **AND-like** in two inducers: low unless both are present. Give the architecture, the state list, the regulation factor, and the truth table it produces.

Then say **which knob you turned** to make it AND-like rather than OR-like, and what you would have to measure to know whether your promoter actually does it.